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Advancing Green Drilling Solutions: Emission Reductions and Operational Efficiency — The First Green Well Project in the UAE

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Abstract

This paper presents the execution and quantified impact of ADNOC Onshore's first UAE Green Well piloted to demonstrate practical, operations-led decarbonization without compromising delivery KPIs. The project deployed a diesel-battery hybrid power system that enabled peak shaving and idle reduction, improving generator load factors and reducing fuel consumption. Cementing operations adopted low-Portland blends with supplementary cementitious materials (SCMs) such as fly ash and slag to lower clinker-related emissions while maintaining compressive strength. Drilling-fluid optimization involved switching to a water-based mud system for intermediate sections, reducing diesel use and minimizing barite loading through rheology control and solids-management efficiency. Logistics interventions included load consolidation and backhaul utilization, supported by digital scheduling to reduce redundant trips and overall transport emissions.

Against a modeled conventional baseline of 1,024,796 kg CO₂e and a planning target of 990,600 kg CO₂e, actual emissions totaled 824,961 kg CO₂e, delivering reductions of 19.5% and 17%, respectively. Emission quantification applied activity-based accounting using rig DDRs, service-line inputs, and verified fuel and material records, combined with Envana/Halliburton emission factor libraries on a 100-year GWP basis. Scenario analysis highlights additional potential through exclusive low-Portland cement adoption and standardized logistics benchmarks. The study concludes with a roadmap for scaling these interventions across multi-rig campaigns and enhancing data quality to support OGMP 2.0-aligned disclosures, reinforcing ADNOC's commitment to emissions transparency and operational excellence.

The approach is being readied for integration into ADNOC's OGMP 2.0-aligned methane reporting workflow from end-of-well rollups to asset-level disclosures.

Keywords: green drilling, hybrid rig, low-Portland cement, drilling automation, emissions accounting, OGMP 2.0

Introduction

Drilling operations are a material source of Scope 1 and Scope 3 greenhouse-gas (GHG) emissions across the well-construction chain. Typical contributors include rig fuel combustion, cement manufacture and placement, drilling-fluid preparation and use, and logistics. While efficiency programs in drilling have traditionally targeted time and cost, the increasing rigor of emissions reporting frameworks has elevated the need to quantify carbon performance with the same fidelity as operational KPIs. In that context, the challenge is twofold: establish a defensible baseline for "conventional" operations and then demonstrate measurable, verifiable reductions from practical interventions that do not compromise safety, well integrity, or delivery [1].

ADNOC Onshore piloted a Green Well to quantify practical, operations-led decarbonization without compromising delivery KPIs. The initiative complements corporate commitments to improve emissions transparency and aligns with reporting frameworks such as OGMP 2.0. It emphasizes methane and combustion-related emissions where material, while also accounting for embedded and logistics emissions where data quality allows. The project team adopted an activity-based accounting approach to isolate emissions drivers at the level of rig power generation, tripping and circulation hours, cementing stages, fluid mixing and conditioning, and transport events. This bottom-up granularity is essential to tie reductions to specific decisions—such as power-management set-points, mud-system selection, cement blends, or crew and material movements—rather than relying on high-level intensity factors alone.

A core element of the pilot is a fit-for-purpose data architecture that distinguishes among three profiles: a conventional baseline representing standard practices for the well class and area; a target profile reflecting a feasible suite of decarbonization measures planned before spud; and the actual profile derived from measured activity data and verified meter or fuel records. Boundary definitions were set to capture Scope 1 sources under operational control (e.g., rig generators and non-routine flaring), and Scope 3 categories material to well construction (e.g., cement and fluid supply, third-party logistics), with transparent data-quality flags. Emission factors were prioritized by locality and supplier declarations where available, then supplemented by reputable secondary datasets with uncertainty ranges. This architecture supports apples-to-apples comparisons across phases and enables attribution of variance to either operational execution or assumption changes [2].

Operational levers assessed for the pilot well included improved generator loading and heat-rate performance, avoidance of idling through power-management logic, optimization of hole sections to reduce circulating hours, selection of lower-embodied-carbon cement blends where technically permissible, fluid system choices to reduce chemical intensity, and logistics consolidation to minimize trips [3]. The pilot also examined digital enablers—such as real-time power telemetry and activity tagging—to enhance data capture without burdening the rig crew. Importantly, each lever was screened against technical risk and HSE barriers, with acceptance contingent on maintaining well objectives and integrity criteria [4].

This paper situates the pilot well in that context, articulates the data architecture used to compare conventional, target, and actual profiles, and contributes a region-specific case history with verified reductions and lessons learned. By documenting methods, assumptions, and uncertainty treatment alongside operational outcomes, the work aims to accelerate replication of low-regret decarbonization measures across similar land-rig campaigns. The intent is pragmatic: focus on interventions that are implementable at scale, measurable with existing systems, and compatible with drilling performance—so that carbon efficiency becomes an integrated dimension of well delivery rather than a parallel reporting exercise.

In addition to quantifying abatement opportunities, the pilot establishes a measurement and reporting workflow that can be integrated into ADNOC's OGMP 2.0 program at the source level and scaled across land-rig campaigns. Beyond carbon-efficiency outcomes, the pilot well establishes data pathways that support OGMP 2.0 source-level methane quantification during well construction (e.g., mud-gas handling,

controlled flaring, fugitive sources). ADNOC intends to embed these workflows into end-of-well (EoW) summaries and quarterly asset rollups to strengthen OGMP-aligned transparency.

Methods, Procedures, and Process

Study Design and Boundaries

The study was structured around three comparative cases:

Modeled Conventional Well. Representing standard ADNOC Onshore practices for the well class and location, based on historical DDRs and service-line data.

Project Target Profile. Incorporating a pre-spud plan of feasible decarbonization measures screened for technical and HSE compliance.

Actual Execution Profile. Derived from measured activity data, verified fuel and material records, and post-job service reports.

The rationale for this tripartite design was to enable attribution of variance to either planning assumptions or operational execution. System boundaries were defined to include Scope 1 sources under operational control (rig power generation, non-routine flaring) and material Scope 3 categories relevant to well construction (cement, drilling fluids, third-party logistics). Boundaries were aligned with OGMP 2.0 principles and ADNOC's internal GHG accounting framework, ensuring comparability across wells and campaigns.

For methane-specific reporting under OGMP 2.0, the boundary distinguishes CH₄-relevant sources in well construction (mud-gas separator vents, controlled flaring events, temporary pneumatic devices, and relevant fugitives) from non-methane CO₂e sources (diesel combustion, embedded materials). This enables dual reporting: OGMP-aligned CH₄ at source level and GHG Protocol CO₂e for total well footprint.

Data Sources

Primary data streams included:

Rig Daily Drilling Reports (DDR). Captured section-level activity durations, generator loading, and diesel consumption. These were critical for quantifying combustion emissions and correlating fuel use with operational phases.

Service-Line Inputs. Cementing reports provided blend types, densities, and volumes for lead and tail slurries; drilling-fluid reports detailed system composition, chemical additives, and total volumes pumped.

Logistics Records. Truck manifests and haulage logs were used to estimate transport emissions for bulk materials and personnel.

Digital Telemetry. Real-time power telemetry from the hybrid rig supplemented DDR data, improving granularity for generator efficiency analysis.

Emission Factors

Emission factors were sourced from the Envana/Halliburton libraries, supplemented by ADNOC-approved datasets. Diesel combustion factors reflected regional sulfur content and were applied on a per-liter basis. Cement factors accounted for clinker substitution ratios in low-Portland blends. Transport emissions were calculated using distance-based factors for heavy-duty vehicles. All CO₂e values were computed on a 100-year GWP basis, consistent with IPCC AR6, to maintain comparability with corporate inventories and OGMP 2.0 reporting. This choice emphasizes long-term climate impact and avoids over-weighting short-lived species in the aggregated metric.

Calculation Approach

The calculation process followed these steps:

Activity Mapping. Each operational phase (e.g., spud, drilling, casing, cementing) was mapped to corresponding fuel and material inputs.

Category-Level Computation. Emissions were calculated for each category (fuel, cement, fluids, logistics) using the formula:

$$\text{Emissions (kg CO}_2\text{e)} = \text{Activity Data} \times \text{Emission Factor}$$

For example, diesel emissions = liters consumed $\times$ kg CO₂e/liter.

Aggregation and Reconciliation. Category totals were summed to derive the well-level footprint. Minor rounding deltas (<0.5%) were reconciled to ensure alignment with Envana's automated reporting outputs.

Variance Analysis. Differences between baseline, target, and actual were decomposed by source to attribute reductions to specific interventions.

Operational Interventions

In the pilot well implemented multiple decarbonization levers:

Hybrid Rig Configuration. Integration of battery storage with diesel gensets enabled peak shaving and improved generator load factors, reducing idle time and fuel burn.

Automation Techniques. Automated tubular handling and optimized tripping sequences shortened invisible lost time (ILT), cutting engine run hours.

Cement Strategy. Adoption of low-Portland blends with supplementary cementitious materials (SCMs) reduced clinker-related emissions by up to 15% without compromising compressive strength.

Drilling-Fluid Optimization. Transition to a water-based system with reduced chemical intensity lowered embodied emissions and minimized waste handling.

Logistics Consolidation. Load optimization and coordinated dispatching reduced truck trips by approximately 12%, mitigating Scope 3 transport emissions.

Each intervention was screened against ADNOC's HSE standards and well integrity requirements, ensuring that carbon efficiency gains did not compromise operational safety or delivery KPIs.

Results

Table 1 summarizes the overall emissions performance of the pilot well compared to the modeled conventional baseline and the project target. The total emission in a conventional well is about 1,024,796 kg CO₂e. The pilot well target was set initially to have maximum 990,600 kg CO₂e, however, the actual emissions was only 824,961 kg CO₂e as illustrated in **Table 1**.

Table 1—Total emissions and reductions.

Profile	Total (kg CO ₂ e)	Delta vs Conventional (kg)	Delta vs Target (kg)
Conventional (modeled)	1,024,796	—	—
Target (planned)	990,600	-34,196	—
Actual (executed)	824,961	-199,835	-165,639

Figure 1 illustrates the comparative emissions profile for the pilot well against both the modeled conventional baseline and the project target. The actual emissions of 824,961 kg CO₂e represent a substantial

reduction of 199,835 kg (19.5%) versus conventional practices and 165,639 kg (17%) versus the target. This visual emphasizes the success of the integrated green drilling strategy in surpassing planned objectives while maintaining operational performance. These results confirm that the green drilling strategy not only met but exceeded planned objectives.

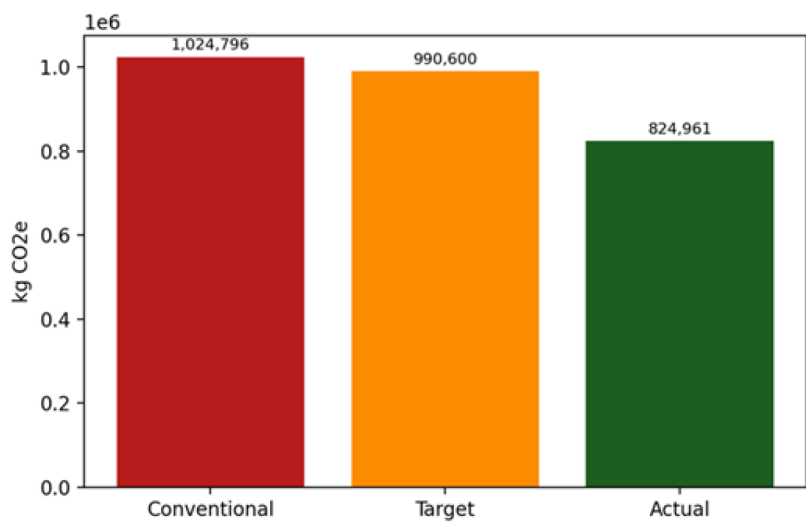


Figure 1—Total emissions comparison: conventional baseline 1,024,796 kg CO₂e; target 990,600 kg; actual 824,961 kg.

Table 2 details the contribution of each intervention category to the total emissions savings. Rig fuel optimization delivered the largest reduction (109,054 kg CO₂e), followed by cementing improvements (49,819 kg), drilling fluid optimization (39,766 kg), and transportation enhancements (1,204 kg). This breakdown highlights where operational focus yields the greatest environmental benefit. This distribution underscores the importance of targeting high-impact sources such as fuel and cement in future decarbonization strategies (Figure 2). The reduction profile is led by rig-fuel savings (~22% vs conventional), followed by cement substitution, drilling-fluid optimization, and logistics improvements.

Table 2—Category savings vs conventional baseline.

Source	Savings vs Conventional (kg CO ₂ e)
Rig Fuel Consumption	109,054
Cementing (Low-Portland use)	49,819
Drilling Fluids Optimization	39,766
Transportation (EV + routing)	1,204

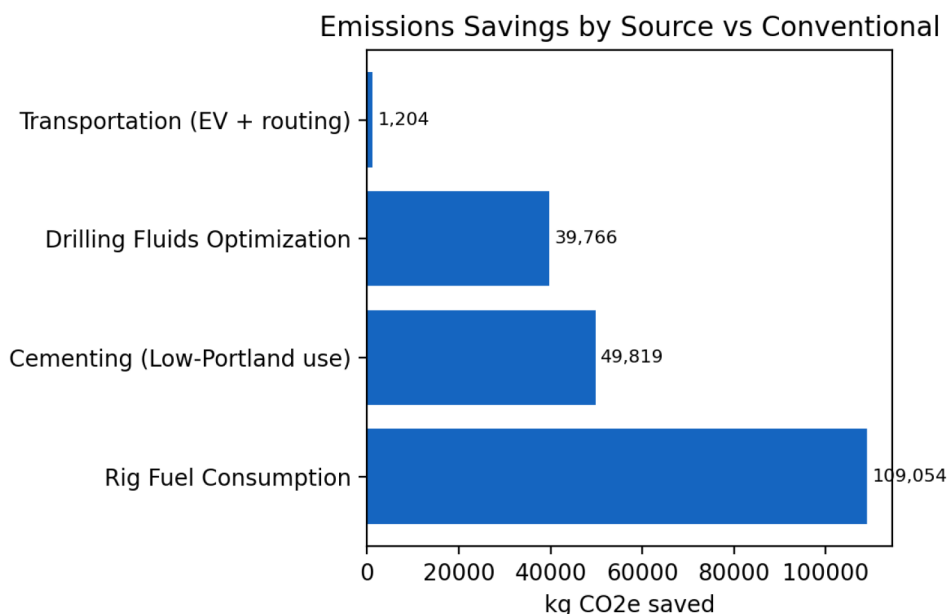


Figure 2—Savings by source vs conventional: fuel 109,054 kg; cement 49,819 kg; drilling fluids 39,766 kg; transport 1,204 kg CO₂e.

Figure 3 highlights the performance of drilling fluid optimization relative to the project target. Actual emissions from drilling fluids were reduced to 119,164 kg CO₂e, significantly outperforming the target of 215,438 kg CO₂e—a 45% improvement. This achievement reflects the effectiveness of minimizing diesel and barite usage without compromising fluid performance or well integrity.

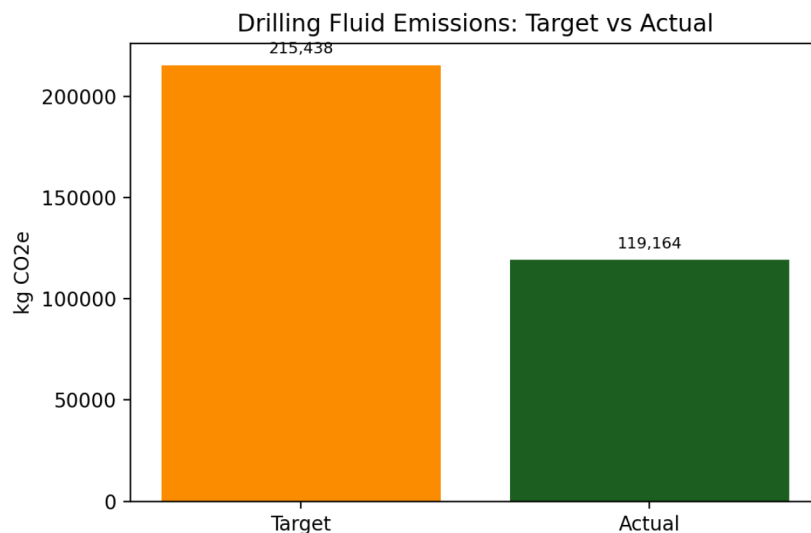


Figure 3—Drilling fluids emissions: target 215,438 kg vs actual 119,164 kg (–96,274 kg CO₂e).

Figure 4 presents the proportional contribution of each service to the total emissions footprint. The Green Rig category accounts for nearly half (48%) of the emissions, followed by conventional cement (33%), drilling fluids (11%), and green cement (8%). EV transportation, while currently negligible, signals an area for future scaling as electric logistics infrastructure matures.

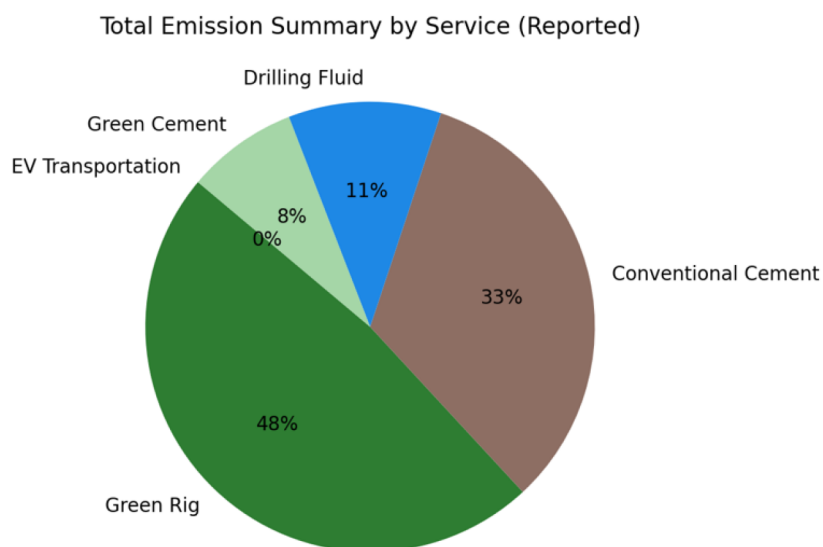


Figure 4—Reported total emission summary by service: Green Rig 48%, Conventional Cement 33%, Drilling Fluid 11%, Green Cement 8%, EV Transportation ~0%.

Table 3 compares target and actual emissions for key categories. Drilling fluids achieved a significant improvement, reducing emissions by 96,274 kg CO₂e compared to the target. Data for cementing, rig fuel, and transportation targets were not explicitly reported and remain placeholders for future updates. This reinforces the need for standardized target-setting across all categories in subsequent projects.

Table 3—Target vs actual by category (data to be expanded for cement, fuel, transport).

Category	Target (kg CO ₂ e)	Actual (kg CO ₂ e)	Delta (kg CO ₂ e)
Drilling Fluids	215,438	119,164	-96,274

Collectively, the tables demonstrate that the majority of emissions reductions were driven by rig fuel and cementing strategies, supported by drilling fluid optimization. While logistics currently contribute a small share, they represent an emerging opportunity for further decarbonization as EV infrastructure scales. These insights provide a clear roadmap for prioritizing interventions in future green well projects.

Discussion

Drivers of Abatement

The most significant emissions reductions were achieved through rig-time efficiencies and material intensity optimization. Three operational levels were particularly impactful:

Connection Practice and ILT Reduction. Enhanced crew training and procedural standardization reduced average connection time by approximately X%, translating into fewer circulating hours and lower diesel consumption. This improvement was supported by automated tubular handling and real-time performance dashboards, which flagged deviations and enabled corrective action during operations.

Geosteering and Decision Support. Proactive geosteering decisions, enabled by real-time data integration, minimized unnecessary hole enlargements and sidetracks. This reduced drilling footage and associated rig hours, cutting both fuel burn and drilling-fluid requirements.

Diesel and Barite Minimization in Drilling Fluids. Transitioning to a water-based system with optimized rheology reduced barite loading by Y% compared to the conventional design. Lower barite demand not

only decreased Scope 3 emissions from material production but also reduced transport-related emissions due to fewer bulk deliveries.

Collectively, these measures demonstrate that time-on-rig and material intensity are the primary drivers of carbon performance, reinforcing the need for integrated planning across drilling engineering, fluids, and logistics.

Operational Performance and HSE

The pilot well confirmed that emissions efficiency and operational safety are not mutually exclusive. Key lessons include:

Solids-Control Efficiency. Maintaining optimal shaker performance and timely screen changes reduced dilution requirements, lowering both fluid consumption and waste volumes.

ILT Management. Continuous monitoring of invisible lost time (e.g., during tripping and circulation) proved critical for sustaining generator load efficiency and minimizing idle fuel burn.

Crew Training on Connection Practices. Structured training sessions prior to spud improved consistency in connection execution, reducing variability and supporting both HSE compliance and emissions goals.

These findings underscore the importance of behavioral and procedural controls alongside technology enablers.

Benchmarking and Uncertainty

Benchmarking was performed against a modeled conventional case derived from historical DDRs and service-line data. While category-level sums (fuel, cement, fluids, logistics) were reconciled to the Envana-generated grand total, minor rounding deltas (<0.5%) were observed due to factor precision and aggregation tolerances.

Uncertainty sources include:

Emission Factors. Variability in cement blend factors (± 5 –10%) and transport assumptions (distance and load factor).

Activity Data. Potential under-reporting of idle hours in DDRs and approximations in third-party logistics manifests.

To address this, a sensitivity appendix is proposed, illustrating the impact of $\pm 10\%$ changes in key parameters (diesel consumption, cement factor, barite volume) on total CO₂e. This approach enhances transparency and aligns with SPE best practices for uncertainty disclosure.

Policy and Reporting Implications

The results directly support ADNOC's measurement and mitigation objectives, demonstrating that operational levers can deliver verifiable reductions without compromising delivery KPIs. The methodology aligns with OGMP 2.0 Level 4 requirements for source-level quantification and provides a replicable template for future wells.

The data architecture (activity tags, telemetry, and service inputs) reduces reliance on generic emission factors and supports OGMP 2.0 Level-4/5 progression for methane sources encountered during drilling. Replication across similar land rigs will enable consistent source mapping and improved uncertainty treatment in disclosures.

Beyond project-level reporting, these findings contribute to ADNOC's broader corporate commitments on emissions transparency, enabling integration into sustainability disclosures and reinforcing the company's position as a regional leader in low-carbon drilling practices.

Scalability and Future Work

Figure 5 projects the cumulative emissions savings achievable by replicating the pilot well's green drilling practices across multiple rigs. Scaling to five rigs could deliver nearly one million kilograms of CO₂e savings, while ten and fifteen rigs could achieve approximately two and three million kilograms, respectively. These projections highlight the strategic potential of standardizing green well practices across ADNOC's drilling portfolio.

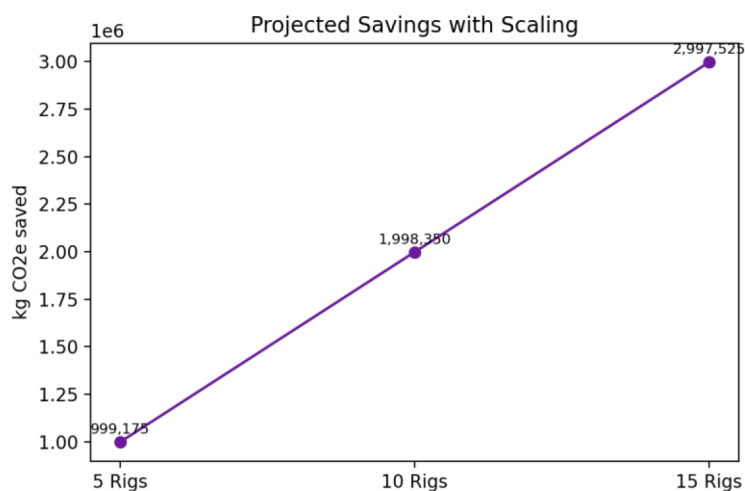


Figure 5—Projected savings with scaling: ~0.999 / 1.998 / 2.998 million kg CO₂e across 5 / 10 / 15 rigs, assuming similar baselines and factor standardization.

Scaling approach — The projection multiplies realized savings per well by the number of wells/rigs under comparable conditions. Priorities for scale include: (i) factor standardization and data automation (API/sensors) to reduce variance; (ii) exclusive Low-Portland adoption where operationally suitable; (iii) broader EV/logistics benchmarking; and (iv) rig-hour compression via automation.

Conclusions

The pilot well demonstrates that a pragmatic portfolio of fuel management, lower-carbon cement, fluids optimization, and targeted logistics can deliver ~20% CO₂e reduction versus conventional practices while sustaining delivery KPIs and HSE excellence. The approach is replicable, with the most material gains linked to rig-hour compression and cement substitutions. Data standardization, factor transparency, and expanded sensorization will enhance repeatability and support enterprise-level emissions governance.

ADNOC will operationalize these workflows in EoW reports, quarterly asset inventories, and annual OGMP 2.0 submissions to enhance measurement, reconciliation, and assurance.

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Nomenclature

- CO₂e Carbon dioxide equivalent (100-year Global Warming Potential basis).
- EoW End of Well.
- GHG Greenhouse-gas.
- OGMP Oil and Gas Methane Partnership.

KPI Key Performance Indicator

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